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Problem 1. , Task 1

If a task isn't on this doc, it's in the code somewhere - either a part of a plot or printed to console.

If we only consider the three nearest-neighbor A-B hoppings, then hopping parameters from a site to itself are taken to be zero, and the Hamiltonian can be expressed as

$$H = t\left(\sum_{i,j} c_i^\dagger c_j + c_j^\dagger c_i\right)$$

where

$$c_i \sim \sum_k e^{ik\cdot r_i} c_k \quad \text{and} \quad c_i^k \sim \sum_k e^{ik\cdot r_i} c_k^\dagger .$$

If we replace j index with a nearest neighbor δ, then the Hamiltonian becomes

$$H = t\left(\sum_{i,\delta} c^\dagger c_{i+\delta} + c_{i+\delta}^\dagger c_i\right)$$

Then expanding in terms of k,

$$H = t\left(\sum_{i,\delta,k,k'} e^{i(k-k')r_i} e^{-ik'\cdot\delta} c_k^\dagger c_{k'} + e^{-i(k-k')r_i} e^{ik'\cdot\delta} c_{k'}^\dagger c_k\right)$$

Since $\sum_i e^{i(k-k')\cdot r_i} = \delta_{kk'}$,

$$H = t\left(\sum_{\delta,k} e^{-ik\cdot\delta} c_k^\dagger c_k + e^{ik\cdot\delta} c_k^\dagger c_k\right)$$

$$= \left(t\sum_{\delta k} e^{-ik\cdot\delta} c_k^\dagger c_k\right) + \left(t\sum e^{+ik\cdot\delta} c_k^\dagger c_k\right)$$

Since $f(k) = \sum_\delta e^{-ik\cdot\delta}$,

$$H(k) = t f(k) + t f^*(k)$$

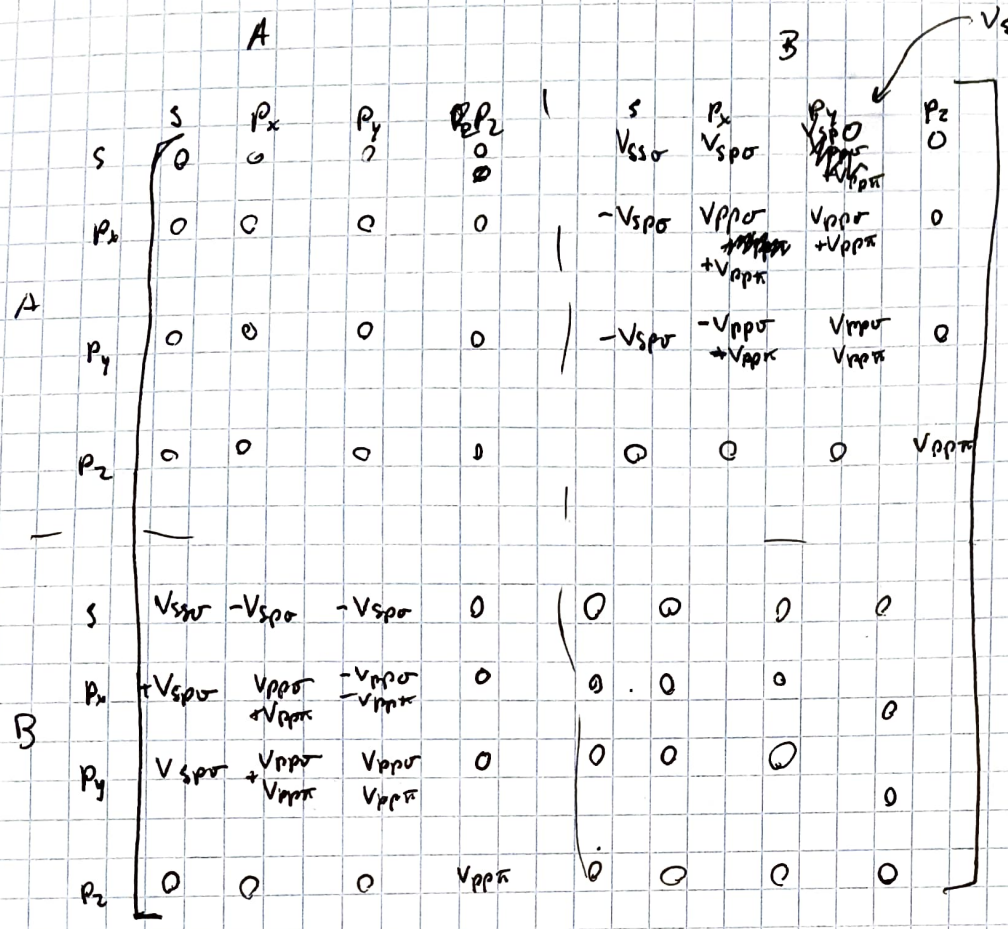
where the two sums are the off-diagonal terms, since the same-site hopping was taken to be zero. Then

$$H(k) = \begin{pmatrix} 0 & t f(k) \\ t f^*(k) & 0 \end{pmatrix}.$$

The orbital interactions involved in the off diagonal elements can be understood through the molecular orbital model.

→

⇒ Carbon has the electron configuration $1s^2 2s^2 2p^2$ with 6 total electrons, so



Where the diagonal blocks have been taken to be zero, and the $p_x p_x$, $p_x p_y$, $p_y p_y$ couplings have components of $V_{pp\sigma}$ and $V_{pp\pi}$ since graphite layers are hexagonal. It's possible that they really only have fractions of each, related to the 120° angles between sites.

Then from the above, the orbital interactions are

$$s_A p_B, \quad s_B p_A, \quad p_A p_B.$$

Task 2

To diagonalize the Bloch Hamiltonian,

$$\hat{H}\vec{v} = E\vec{v}$$

$$\Rightarrow (H - EI)\vec{v} = 0$$

$$\Rightarrow \begin{pmatrix} 0-E & t f(k) \\ t f^*(k) & 0-E \end{pmatrix}\vec{v} = 0 \qquad \det\begin{pmatrix} -E & t f(k) \\ t f^*(k) & -E \end{pmatrix} \overset{\text{set}}{=} 0$$

$$\Rightarrow \begin{pmatrix} -E & t f(k) \\ t f^*(k) & -E \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad \Rightarrow E^2 - t^2|f(k)|^2 = 0$$

$$\Rightarrow E_{\pm} = \pm t|f(k)|$$

$$\Rightarrow -E v_1 + t f(k) v_2 = 0 \Rightarrow v_1 = \frac{t f(k)}{E} v_2$$

$$t f^*(k) v_1 - E v_2 = 0$$

$$\hookrightarrow \vec{v} = \begin{pmatrix} t f(k)/E \\ 1 \end{pmatrix} \rightarrow$$

$$\rightarrow \vec{v}_1 = \begin{pmatrix} f(k)/|f(k)| \\ 1 \end{pmatrix}, \quad \vec{v}_2 = \begin{pmatrix} -f(k)/|f(k)| \\ 1 \end{pmatrix}$$

$$V = \begin{pmatrix} -\frac{f(k)}{|f(k)|} & \frac{f(k)}{|f(k)|} \\ 1 & 1 \end{pmatrix}, \quad V^{-1} = \frac{-1}{2f(k)/|f(k)|} \begin{pmatrix} -1 & f(k)/|f(k)| \\ -1 & f(k)/|f(k)| \end{pmatrix}$$

$$D = V^{-1} H V$$

$$= \frac{+|f(k)|}{2f(k)} \begin{pmatrix} -1 & +f(k)/|f(k)| \\ +1 & +f(k)/|f(k)| \end{pmatrix} \begin{pmatrix} 0 & tf(k) \\ tf^*(k) & 0 \end{pmatrix} \begin{pmatrix} -\frac{f(k)}{|f(k)|} & \frac{f(k)}{|f(k)|} \\ 1 & 1 \end{pmatrix}$$

$$= \frac{+|f(k)|}{2f(k)} \begin{pmatrix} +t & -tf(k) \\ +t & +tf(k) \end{pmatrix} \begin{pmatrix} -\frac{f(k)}{|f(k)|} & \frac{f(k)}{|f(k)|} \\ 1 & 1 \end{pmatrix}$$

$$= \frac{-(f(k)|}{2f(k)} \begin{pmatrix} tf(k)/|f(k)| + tf(k) & \\ & \end{pmatrix}$$

$$= \frac{|f(k)|}{2f(k)} \begin{pmatrix} t|f(k)| & -tf(k) \\ t|f(k)| & tf(k) \end{pmatrix} \begin{pmatrix} -f(k)/|f(k)| & f(k)/|f(k)| \\ 1 & 1 \end{pmatrix}$$

$$= \frac{|f(k)|}{2f(k)} \begin{pmatrix} -tf(k) - tf(k) & 0 \\ 0 & tf(k) + tf(k) \end{pmatrix} = \frac{|f(k)|}{2f(k)} \begin{pmatrix} -2tf(k) & 0 \\ 0 & tf(k)+tf(k) \end{pmatrix}$$

with

$$= \begin{pmatrix} -t|f(k)| & 0 \\ 0 & tf(k) \end{pmatrix} = \begin{pmatrix} +E_1 & 0 \\ 0 & E_2 \end{pmatrix}$$

Task 6

One p_z orbital in each site in the primitive cell because the p_z orbitals are somewhat decoupled from the rest of the orbitals, as graphite's carbon atoms experience sp^2 hybridization (which doesn't really happen, as we learned in class, but it gives the right shapes and is good enough). The in-plane p-orbitals and the s-orbital behave as though they're filled by the other carbon sites in the layer as one s-orbital electron is promoted to the p_z orbital. Then, since there are two electrons between the two inequivalent sites that are available to hop easily, the band structure shows two bands as π-bands. A more detailed model would show bands from bonds formed between other orbitals, but the electron transport properties of graphite are largely due to the p_z orbitals anyway.

Half-filling places the Fermi level right at the band touching since the eigenvectors are the same on one coordinate and opposite in the other while their eigenvalues are negatives of each other. This allows for the symmetric band structure about the Fermi energy line. The Fermi energy is zero since the difference in on-site energies is zero, so half of this is zero, the value of the Fermi level. Filling half way, filling the valence band fills energy levels up to zero, and since the difference in on-site energies is zero,

this places the band touching at the Fermi level.

Problem 2

Task 6

Since the p_z electrons are the only ones that really do any meaningful hopping, and we consider hoppings between different sites, there will be a total of four bands for the four electrons, one per site, two per layer, and the diagonal elements are still zero, as a level to set the zero-energy. The other zero-elements are zero since they involve hopping from an A-site in one layer to an A-site in the other, from a B-site in one layer to a B-site in the other, or from an A-site/B-site in one layer to another A-site/B-site in the same layer. Only neighboring inequivalent site hoppings are considered, so the Bloch construction is the same. Within each block, we can also see diagonal elements of zero in each, following the rules for building the Hamiltonian.

Task 5

K-point eigenvectors are classified as Dimer and non-Dimer in the code and printed to the console, as are the eigenvalues and the weights for each eigenvector.

The Hamiltonian is in general

$$\hat{H} = \begin{pmatrix} 0 & tf(k) & 0 & 0 \\ tf^*(k) & 0 & \gamma_1 & 0 \\ 0 & \gamma_1 & 0 & tf(k) \\ 0 & tf^*(k) & 0 & 0 \end{pmatrix}$$

@ K, $k = (4\pi/3a, 0)$

$$\rightarrow f(k) = e^{i\left(\frac{4\pi}{3a}\right)(0)} + e^{i\left(\frac{4\pi}{3a}\right)\left(\frac{a}{2}\right)} + e^{-i\left(\frac{4\pi}{3a}\right)\left(\frac{a}{2}\right)}$$

$$= 1 + 2\cos\left(\frac{2\pi}{3}\right) = 1 + 2\left(-\tfrac{1}{2}\right) = 1 - 1 = 0$$

$$\rightarrow \hat{H}_K = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \gamma_1 & 0 \\ 0 & \gamma_1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & \Gamma_1 \\ \Gamma_1 & 0 \end{pmatrix}$$

So the energies at K are not zero.

$$\hat{H} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \gamma_1 & 0 \\ 0 & \gamma_1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow (H - EI) = \begin{pmatrix} -E & 0 & 0 & 0 \\ 0 & -E & \gamma_1 & 0 \\ 0 & \gamma_1 & -E & 0 \\ 0 & 0 & 0 & -E \end{pmatrix}$$

$$\rightarrow \det(H - EI) = -E \begin{vmatrix} -E & \gamma_1 & 0 \\ \gamma_1 & -E & 0 \\ 0 & 0 & -E \end{vmatrix} = -E\left[-E(E^2) - \gamma_1(-\gamma_1 E)\right] = 0$$

$$\rightarrow E^4 - \gamma_1^2 E^2 = 0 = E^2(E^2 - \gamma_1^2) = 0$$

$$\rightarrow E = 0, 0, \pm\gamma_1$$

So there's a degenerate eigenvalue of zero ~~=~~ the band touching ~~=~~ and two split eigenvalues $-\gamma_1$ and γ_1. The zero-energies are expected from the single-layer model, and the dimer splitting energies come from the vertical hopping allowed by the second layer.

Problem 3

Task 1

There are two nearby layers that the (for example) B_1 orbital can hop to: an A_2 site above, and an A_2 site below. Consider the ~~Fourier transform~~ hopping sum for the vertical hops:

$$g(\vec{k}_z) = \gamma_1 \sum_{\delta_z} e^{i\vec{k}_z \cdot \delta_z}$$

where I use δ_z to represent nearest vertical neighbors. Since we have periodicity, the layers are equally spaced with inequivalent sites placed directly over each other, making an alternating pattern. But this forces $\vec{k}_z \cdot \delta_z$ to be $|k_z||\delta_z| = k_z c/2$, where $c/2$ is the vertical spacing, so long as we recognize this δ_z to allow the electron to jump up a layer. A jump down would produce the same value for the dot product but with a minus sign since the vertical nearest neighbors are directly above/below inequivalent sites. Other components to the sum are disregarded as they aren't nearest neighbors, and we're only considering nearest neighbors, so the sum becomes

$$\begin{aligned} g(\vec{k}_z) &= \gamma_1 \sum_{\delta_z} e^{i\vec{k}_z \cdot \delta_z} \\ &= \gamma_1 \left(e^{ik_z c} + e^{-ik_z c} \right) \\ &= \gamma_1 e^{ik_z c/2} + \gamma_1 e^{-ik_z c/2} \\ &= 2\gamma_1 \cos(k_z c/2) \end{aligned}$$

Task 3

The limiting forms are verified by the code with results in the console. The vertical hopping function has an absolute range of

$$0 \leq 2\gamma_1 \left| \cos(k_z c/2) \right| \leq 2\gamma_1 .$$

The minimum coupling / maximum decoupling, when $g(k_z) = 0$, happens when $k_z = \pi/c$

$$g(\pi/c) = 2\gamma_1 \cos\left(\frac{\pi c}{c2}\right) = 2\gamma_1 \cos(\pi/2) = 2\gamma_1 (0) = 0$$

and the maximum coupling, when $g(k_z) = 2\gamma_1$, happens when $k_z = 0$

$$g(0) = 2\gamma_1 \cos(0 \cdot c/2) = 2\gamma_1 \cos(0) = 2\gamma_1 (1) = 2\gamma_1$$

When $k_z = \pi/c$, the dimer sites are completely out of phase, vertically. When $k_z = 0$, the dimer sites are completely in phase, vertically. When $k_z = \pi/2$, the forward and backward ~~phases~~ (layer-to-layer) phases are opposite with an effective period of $\pi/2$, while when $k_z = 0$, the forward and backward phases are the same with an effective period of zero for max coupling.

Task 5

Harrison's Rule is usually used for stronger covalent-like bonds. The connection between dimer sites in vertical graphite stacking isn't covalent. According to Wikipedia, the bonds between layers are van der Waals bonds, which are much weaker, and allow for pencils to write on paper. Since these bonds are weak though, an appropriate parameter should be used to more accurately model the properties of graphite. The empirical value does this. Also, it's empirical, so of course it matches up better with real graphite. The Harrison $V_{pp\sigma}$ parameter is estimated in the code and printed to the console. It's about 1.51 eV, which is over three times the empirical value of 0.390 eV. The resulting bands show wider splitting and less energetic bands in general. The DOS is also very triangular rather than somewhat trapezoidal in appearance, but the state counting and 0 DOS at the Fermi level are maintained.

Task 6

The model predicts that graphite is semi-metallic. At the K and H points, the conduction and valence bands meet at the Fermi level with zero DOS. It predicts the DOS is symmetric about the $E=0$ line, and is based only on p_z-p_z hopping.

To get a quantitatively accurate Fermi surface for graphite, hopping parameters between layers for non-dimer sites should be included as well as hopping between non-identical, non-nearest-neighbor sites within layers.